

Akshat Anand

Software Engineer | Full-Stack Systems | AI Platforms | Real-Time Architecture
Pune, Maharashtra | toakshh@gmail.com | [+91 9860997478](tel:+919860997478) | github.com/toakshh | toakshh-portfolio.vercel.app

Professional Summary

Built and shipped production AI platforms, real-time SDKs, and multi-provider LLM systems across multiple personal and professional projects — TypeScript, React, Node.js, Docker, LangChain, GPT-4o, Gemini, Pinecone, Socket.IO. Getting to 3 years of end-to-end ownership across frontend, backend, AI, and SDK layers, from tokenizer internals through containerised production deployment.

Technical Skills

- **Frontend:** React, Next.js, Three.js, TypeScript, Redux, Zustand, Tailwind CSS, Framer Motion, GSAP
- **Backend:** Node.js, Express.js, MongoDB, PostgreSQL, REST APIs, JWT Authentication
- **AI & Data:** LLMs, LangChain, RAG pipelines, OpenAI (GPT-4o), Google Gemini, Pinecone, Vector Databases, TTS, Embeddings, Prompt Engineering
- **DevOps & SDK:** Docker, CI/CD (GitHub Actions), Webpack, TypeScript library architecture
- **Real-Time:** Streaming architectures, Socket.IO, event-driven systems, WebSocket, multi-provider LLM + TTS pipelines
- **Cross-Platform & Architecture:** React Native, Expo, Electron, Prisma, Supabase, Zustand, microservices, JIRA

Experience

Web Developer — Metabrix Lab Aug 2025 – Present

- **40% faster dashboard load** — replaced a monolithic polling architecture with event-driven Node.js microservices containerised in Docker, cutting mean time to insight across 3 production dashboards.
- **Zero-downtime deploys** across 10+ releases — introduced GitHub Actions CI/CD with automated test gates and Docker image versioning, eliminating manual deployment errors entirely.
- **30% sprint throughput increase** in two Agile cycles — led system design decisions and rolled out modular frontend architecture standards adopted across the full engineering team.
- **25% drop in analyst review time** — built a multi-provider LLM + TTS system (OpenAI, Gemini) with real-time streaming and comprehensive latency metrics, integrated via LangChain into production dashboards without page reloads.

Freelance Software Engineer — Independent Mar 2025 – Jul 2025

- **AI document assistant** (OpenAI + PostgreSQL + React) handling 100+ daily queries, reducing client manual processing by 3 hours/day — delivered solo in 6 weeks.
- **3 client APIs** containerised and deployed on DigitalOcean using Docker, with JWT-secured auth and MongoDB/PostgreSQL storage, serving paying clients with zero reported auth failures.
- **4 full-stack systems** shipped across web and mobile in 5 months — each on time, using Agile sprint planning and reusable component libraries that cut average setup time by 35%.

Frontend Developer — Zimension 3D Technologies Mar 2024 – Dec 2024

- **45% FPS improvement** in browser-based 3D scenes — redesigned the Three.js rendering pipeline with geometry instancing, level-of-detail switching, and texture atlasing (verified via Chrome DevTools).
- **CVEs reduced from 3 to 0** — overhauled access control and input sanitisation across the application, verified through an internal security audit.
- **60% fewer React re-renders** — refactored state management to Redux with memoised selectors (measured via React Profiler), directly improving responsiveness on low-powered devices.
- **3 production releases**, zero rollbacks over 9 months — shipping in a 5-person Agile team using JIRA for sprint planning and code reviews.

Notable Projects

Ultron AI Ecosystem

- Official JS/TS SDK (Rollup-bundled, npm-published) for real-time AI vision, screen commentary, session management, and TTS, with WebSocket/Socket.IO transport, JWT auth, streaming interfaces across React, and Electron, all containerised with Docker and shipped via GitHub Actions CI/CD.

AI RAG & Data Pipeline

- Context-aware AI chatbot backend with Pinecone vector store, LangChain RAG orchestration, and OpenAI embeddings — JWT-secured Express API on MongoDB, delivering 50% faster semantic retrieval vs. keyword baselines.

Video Frame Analyzer SDK

- Real-time video frame analysis SDK built on WebAssembly (WASM) for perceptual hashing and SSIM comparison — TypeScript-first, with a full build pipeline (build:wasm, build:all) and browser-compatible demo harness.

Certifications & Education

Full Stack Fellowship — Crio.do (2023)

GenAI Cohort — Chai aur Code (2025)

BA, English Literature — L.N.M University